

$\text{Aut}(F_2)$ and ACA Equivalence

A Formal Tutorial and Practice Workbook

Why 261 benchmark presentations reduce to at most 124 distinct problems

ACSolverX equivalence-class analysis

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Central claim. The computation does not prove that any of the 261 presentations is AC-trivial. It proves that a finite graph of certified Andrews–Curtis and change-of-basis edges on the 261 inputs has 124 connected components. Consequently, proving one representative from each component AC-trivial would prove all 261 AC-trivial. Each verified component is contained in a true ACA class, so 124 is an upper bound on the number of distinct problems, not a proof that 124 mutually inequivalent AC-classes exist.

Contents

How to use this guide	1
1 Conventions: three elementary AC moves	1
2 Free groups and balanced presentations	2
2.1 The free group F_2	2
2.2 Presentations are quotients of the free group	3
3 Automorphisms of F_2 and their action on presentations	4
3.1 Why the presented group is unchanged	5
3.2 How the project verifies an automorphism	5
4 Transporting AC moves through a fixed automorphism	6
4.1 Quantitative bounds	6
4.2 From the transformed endpoint back to (x, y)	7
5 Canonicalization and the eight signed relabelings	9
5.1 Canonicalization	9
5.2 The eight signed permutations	10
5.3 Worked example: 19_45 and 19_50	10
5.4 Counts	11
6 Full Aut(F_2) and Whitehead canonicalization	11
6.1 The twenty elementary Whitehead generators	11
6.2 The classical theorem used	12
6.3 Independent checks	13
6.4 Empirical Aut witnesses on the 93 pure change-of-variable edges	13
6.5 Worked example: 19_37 and 17_41	13
6.6 Exact count	14
7 Definition 2.1 S-moves as elementary AC sequences	14
7.1 Rotations are conjugates	15
7.2 A constructive elementary realization	15
7.3 Canonicalization cost	16
7.4 Transporting an S-move through ϕ	16

8	ACA: the equivalence relation used by the search	17
8.1	The quotient search graph	17
8.2	Soundness and incompleteness	17
9	The complete reduction: 261 to at most 124	18
9.1	From 168 to 126	18
9.2	Why 126 became 125	19
9.3	Why 125 became 124	19
9.4	The coverage theorem	19
9.5	How to obtain a 124-representative workload	20
9.6	What 124 does and does not mean	20
10	Certificates: why the mergers are proofs	21
10.1	Two edge types	21
10.2	Independent verification	21
10.3	Mutation testing	22
10.4	Invariant cross-check	22
11	Presentation-ready synthesis	23
11.1	A concise theorem chain	23
11.2	Move-bound summary	23
11.3	Claims to make	23
11.4	Claims not to make	23
11.5	Oral rehearsal questions	24
12	Complete answer key and learning diagnostics	25
13	Sources and reproducibility map	37

How to use this guide

This document has three simultaneous purposes.

1. It develops the group theory needed to understand why $\text{Aut}(F_2)$ is relevant to balanced presentations.
2. It proves, move by move, why a change of free basis preserves AC-triviality and gives quantitative upper bounds on the transported number of elementary AC moves.
3. It explains the complete computational pipeline

$$261 \longrightarrow 261 \longrightarrow 171 \longrightarrow 168 \longrightarrow 126 \longrightarrow 125 \longrightarrow 124$$

and separates exact conclusions from search-dependent upper bounds.

Every substantive section ends with exercises. A complete answer key, including diagnostic advice, begins in Section 12. The final summary in Section 11 is designed for a lab presentation.

Warning 0.1 (No silent theorem dependencies). Elementary free-group and AC facts are proved here. Whitehead peak reduction and the Nielsen basis theorem are classical results too large to reprove honestly in a tutorial of this scope. They are stated with their hypotheses, their precise role is isolated, and every project-specific conclusion drawn from them is proved.

1 Conventions: three elementary AC moves

There are multiple numbering conventions in the literature and in the ACSolverX materials. Mixing them changes move counts. This guide uses the convention of the source paper `literature/txt/math_ml_paper_2408.15332.txt`, lines 296–298.

Definition 1.1 (Source-paper elementary AC moves). Let

$$P = \langle x_1, \dots, x_n \mid r_1, \dots, r_n \rangle$$

be balanced. For $i \neq j$, the elementary moves are:

$$\begin{aligned} \text{(AC1)} \quad & r_i \longmapsto r_i r_j, \\ \text{(AC2)} \quad & r_i \longmapsto r_i^{-1}, \\ \text{(AC3)} \quad & r_i \longmapsto g r_i g^{-1}, \quad g \in \{x_1^{\pm 1}, \dots, x_n^{\pm 1}\}. \end{aligned}$$

Only one relator is replaced; all other relators remain unchanged.

Operation	Source-paper convention used here	Two-Hump/solver prose convention
Combine	AC1: $r_i \mapsto r_i r_j$	Often called AC2
Invert	AC2: $r_i \mapsto r_i^{-1}$	Often called AC1
Conjugate	AC3 by one generator or inverse	Often AC3 by an arbitrary word
Solver S-move	A finite sequence of AC1–AC3	One encoded macro-move

Warning 1.2 (No inverse multiplier in elementary AC1). The source-paper AC1 is $r_i \mapsto r_i r_j$, not $r_i \mapsto r_i r_j^{-1}$. The latter can be realized while restoring r_j by three moves:

$$(r_i, r_j) \xrightarrow{\text{AC2}} (r_i, r_j^{-1}) \xrightarrow{\text{AC1}} (r_i r_j^{-1}, r_j^{-1}) \xrightarrow{\text{AC2}} (r_i r_j^{-1}, r_j).$$

Thus the solver parameter $j = -1$ is part of a macro-move under the source-paper convention.

Relator order is not mathematical data, so swapping r_1, r_2 is treated as zero-cost bookkeeping. Free reduction is also zero-cost because it replaces a word by the same element of the free group.

Exercises

Exercise 1.1. Using the convention above, write a three-move sequence that reverses one positive AC1 move:

$$(r_1 r_2, r_2) \longrightarrow (r_1, r_2).$$

Exercise 1.2. Explain why the phrase “one rotation move” is ambiguous unless one states whether AC3 allows conjugation by an arbitrary word or only by one generator.

Exercise 1.3. Classify each operation as elementary, macro, or bookkeeping:

$$r_1 \mapsto r_1^{-1}, \quad r_1 \mapsto r_1 r_2^{-1}, \quad r_1 \mapsto w r_1 w^{-1}, \quad (r_1, r_2) \mapsto (r_2, r_1).$$

Use the source-paper convention.

2 Free groups and balanced presentations

2.1 The free group F_2

Definition 2.1 (Free group on two generators). The free group

$$F_2 = \langle x, y \rangle$$

consists of reduced words in the alphabet

$$\{x, X, y, Y\}, \quad X = x^{-1}, \quad Y = y^{-1}.$$

A word is reduced when it contains no adjacent inverse pair xx, Xx, yY, Yy . Multiplication is concatenation followed by free reduction.

For a word $w = a_1 \cdots a_m$, its inverse is

$$w^{-1} = a_m^{-1} \cdots a_1^{-1}.$$

Thus, for example,

$$(xyX)^{-1} = xYX.$$

The empty word is the identity. We write $|w|$ for the length of the freely reduced word representing w .

Theorem 2.2 (Universal property of F_2). *Let H be any group and choose $a, b \in H$. There exists a unique homomorphism $f : F_2 \rightarrow H$ satisfying $f(x) = a$ and $f(y) = b$.*

Proof. Define f on a word by replacing each letter:

$$x \mapsto a, \quad X \mapsto a^{-1}, \quad y \mapsto b, \quad Y \mapsto b^{-1},$$

and multiplying the resulting elements of H . Deleting an adjacent inverse pair in F_2 deletes aa^{-1} , $a^{-1}a$, bb^{-1} , or $b^{-1}b$ in H , so the value is unchanged by free reduction. Therefore the map is well-defined. Concatenation gives $f(uv) = f(u)f(v)$, so it is a homomorphism. Any homomorphism with the specified values on x, y must evaluate every word by the same rule, which proves uniqueness. $\square$

This theorem explains why assigning words to x and y automatically defines an endomorphism $F_2 \rightarrow F_2$. It does *not* automatically define an automorphism: invertibility must still be proved.

2.2 Presentations are quotients of the free group

Definition 2.3 (Normal closure). For $r_1, r_2 \in F_2$, the normal closure

$$\langle\langle r_1, r_2 \rangle\rangle$$

is the smallest normal subgroup of F_2 containing r_1, r_2 . Equivalently, its elements are finite products of conjugates of $r_1^{\pm 1}$ and $r_2^{\pm 1}$.

Definition 2.4 (Balanced two-generator presentation).

$$P = \langle x, y \mid r_1, r_2 \rangle$$

denotes the quotient group

$$G_P = F_2 / \langle\langle r_1, r_2 \rangle\rangle.$$

It is balanced because the number of relators equals the number of generators. Balanced does not mean trivial.

The presentation defines the trivial group precisely when

$$G_P = 1 \iff \langle\langle r_1, r_2 \rangle\rangle = F_2.$$

The standard trivial presentation is

$$T = \langle x, y \mid x, y \rangle.$$

Definition 2.5 (AC-trivial). A balanced presentation is AC-trivial when a finite sequence of elementary AC moves carries its relator pair to the standard pair (x, y) , up to free reduction and relator order.

Warning 2.6 (Trivial group versus AC-trivial presentation). If a presentation is AC-trivial, it certainly presents the trivial group. The converse is the Andrews–Curtis conjecture and is not known in general. The 261 benchmark inputs present the trivial group, but none has been proved AC-trivial by the equivalence-class search.

Exercises

Exercise 2.1. Freely reduce $xyYXxXY$, and compute the inverse of the reduced word.

Exercise 2.2. Prove directly that

$$\langle x, y \mid xy, y \rangle$$

presents the trivial group. Then give a source-convention AC sequence to (x, y) .

Exercise 2.3. Why does the equality $G_P = 1$ not by itself provide an AC path from (r_1, r_2) to (x, y) ?

Exercise 2.4. State exactly where balancedness is used in the definition of the Andrews–Curtis problem, and explain why a two-generator presentation with three relators is not one of the objects counted in the 261-to-124 pipeline.

3 Automorphisms of F_2 and their action on presentations

Definition 3.1. An endomorphism $\phi : F_2 \rightarrow F_2$ is an automorphism if it is bijective. The group of all such maps under composition is $\text{Aut}(F_2)$.

By Theorem 2.2, words $u, v \in F_2$ determine a unique endomorphism

$$\phi(x) = u, \quad \phi(y) = v.$$

To prove that ϕ is an automorphism, it is enough to construct an endomorphism ψ satisfying both

$$\psi \circ \phi = \text{id}_{F_2}, \quad \phi \circ \psi = \text{id}_{F_2}.$$

Because homomorphisms out of F_2 are determined by x, y , each equality needs to be checked only on those two generators.

Example 3.2 (The basic length-changing automorphism). Let

$$\phi(x) = xy, \quad \phi(y) = y.$$

The universal property makes this a homomorphism. Define

$$\psi(x) = xY, \quad \psi(y) = y.$$

Then

$$\begin{aligned} \psi(\phi(x)) &= \psi(xy) = (xY)y = x, & \psi(\phi(y)) &= y, \\ \phi(\psi(x)) &= \phi(xY) = (xy)Y = x, & \phi(\psi(y)) &= y. \end{aligned}$$

Hence $\psi = \phi^{-1}$, so $\phi \in \text{Aut}(F_2)$.

Remark 3.3. The rule on inverse letters is forced:

$$\phi(X) = \phi(x^{-1}) = \phi(x)^{-1}, \quad \phi(Y) = \phi(y)^{-1}.$$

For Example 3.2,

$$\phi(X) = (xy)^{-1} = YX, \quad \phi(Y) = Y.$$

One must reverse the image word and invert every letter.

3.1 Why the presented group is unchanged

For $P = \langle x, y \mid r_1, r_2 \rangle$, define

$$\phi(P) = \langle x, y \mid \phi(r_1), \phi(r_2) \rangle.$$

The symbols x, y remain the displayed free generators; the relator words have been rewritten in a new free basis.

Theorem 3.4 (Change of free basis induces a quotient isomorphism). *Let $N = \langle\langle r_1, r_2 \rangle\rangle$. If $\phi \in \text{Aut}(F_2)$, then*

$$\phi(N) = \langle\langle \phi(r_1), \phi(r_2) \rangle\rangle$$

and

$$\bar{\phi} : F_2/N \longrightarrow F_2/\phi(N), \quad wN \longmapsto \phi(w)\phi(N)$$

is an isomorphism.

Proof. An automorphism sends products, inverses, and conjugates to products, inverses, and conjugates. Therefore $\phi(N)$ is a normal subgroup containing $\phi(r_1), \phi(r_2)$, so $\langle\langle \phi(r_1), \phi(r_2) \rangle\rangle \subseteq \phi(N)$. Applying the same argument to ϕ^{-1} gives the reverse inclusion.

If $wN = w'N$, then $w^{-1}w' \in N$, hence $\phi(w)^{-1}\phi(w') \in \phi(N)$, so $\bar{\phi}$ is well-defined. It is a homomorphism because ϕ is. The map induced by ϕ^{-1} is its two-sided inverse. $\square$

Corollary 3.5. *If P presents the trivial group, then $\phi(P)$ presents the trivial group. The transformed presentation remains balanced.*

Warning 3.6. Theorem 3.4 proves that P and $\phi(P)$ present isomorphic groups. That alone does not prove either presentation AC-trivial. The stronger AC-triviality statement requires the transported path theorem in Section 4.

3.2 How the project verifies an automorphism

The equivalence pipeline does not trust an arbitrary dictionary $\{x \mapsto u, y \mapsto v\}$. It independently checks that (u, v) is a free basis using Nielsen reduction. The classical Nielsen basis theorem says that a generating pair is a basis exactly when elementary Nielsen transformations reduce it to a signed permutation of (x, y) . The verifier in `experiments/equivalence_classes/verify/verify_certificates.py` implements this check independently of the Whitehead code that produced the witness.

Exercises

Exercise 3.1. Prove that

$$\phi_k(x) = xy^k, \quad \phi_k(y) = y$$

is an automorphism for every $k \in \mathbb{Z}$, by constructing its inverse.

Exercise 3.2. Apply $\phi(x) = xy, \phi(y) = y$ to the word $YXyX$, showing the image of each capital letter before reducing.

Exercise 3.3. Let $\eta(x) = x^2, \eta(y) = y$. Explain why the universal property proves that η is a homomorphism but not that it is an automorphism. Prove that it is not surjective by using exponent sums.

Exercise 3.4. Prove Theorem 3.4 in the special case $P = \langle x, y \mid x, y \rangle$ and $\phi(x) = xy, \phi(y) = y$, without invoking the general proof.

4 Transporting AC moves through a fixed automorphism

This is the central mathematical point. The correct statement is about reachability, not necessarily adjacency.

Theorem 4.1 (Transported elementary move). *Let $\phi \in \text{Aut}(F_2)$. If*

$$(r_1, r_2) \xrightarrow{\text{one elementary AC move}} (r'_1, r'_2),$$

then

$$(\phi(r_1), \phi(r_2)) \xrightarrow{\text{AC}^*} (\phi(r'_1), \phi(r'_2)),$$

where AC^ denotes a finite sequence of source-convention elementary AC moves.*

Proof. We inspect all three elementary moves.

AC1. If $r'_i = r_i r_j$, then

$$\phi(r'_i) = \phi(r_i r_j) = \phi(r_i) \phi(r_j).$$

Thus one transformed AC1 move suffices.

AC2. If $r'_i = r_i^{-1}$, then

$$\phi(r'_i) = \phi(r_i^{-1}) = \phi(r_i)^{-1}.$$

Thus one transformed AC2 move suffices.

AC3. Suppose $r'_i = g r_i g^{-1}$, where g is one generator or inverse. Write the freely reduced word

$$\phi(g) = a_1 a_2 \cdots a_m, \quad a_s \in \{x, X, y, Y\}.$$

Starting from $R = \phi(r_i)$, conjugate successively by $a_m, a_{m-1}, \dots, a_1$. After m source-convention AC3 moves, the result is

$$a_1 \cdots a_m R a_m^{-1} \cdots a_1^{-1} = \phi(g) \phi(r_i) \phi(g)^{-1} = \phi(r'_i).$$

The other relator is untouched throughout. □

4.1 Quantitative bounds

Definition 4.2. For a fixed automorphism ϕ , define its generator expansion factor

$$\Lambda(\phi) = \max\{|\phi(x)|, |\phi(y)|\}.$$

Since inversion preserves reduced length, $|\phi(X)| = |\phi(x)|$ and $|\phi(Y)| = |\phi(y)|$.

Proposition 4.3 (Per-move transport bound). *Under the source-paper convention, the constructive transported costs are:*

<i>original elementary move</i>	<i>number of transformed elementary moves</i>
AC1	1
AC2	1
AC3 by g	$ \phi(g) \leq \Lambda(\phi)$.

These are guaranteed upper bounds for the displayed construction; they are not claims about shortest AC distance.

Corollary 4.4 (Whole-path bound). *Suppose an AC path contains N_1 AC1 moves, N_2 AC2 moves, and AC3 conjugations by letters $g_1, \dots, g_{N_3}$. Transporting the path through ϕ gives a path of length at most*

$$N_1 + N_2 + \sum_{t=1}^{N_3} |\phi(g_t)| \leq N_1 + N_2 + \Lambda(\phi)N_3.$$

If $N = N_1 + N_2 + N_3$, then the coarser bound is

$$N_\phi \leq \Lambda(\phi)N.$$

The last inequality uses $\Lambda(\phi) \geq 1$, which holds because an automorphism cannot send a free generator to the identity.

Example 4.5. For $\phi(x) = xy, \phi(y) = y$, one has $\Lambda(\phi) = 2$. Conjugation by x or X costs at most two transformed AC3 moves; conjugation by y or Y costs one. Hence a path of length N transports to one of length at most $2N$, before correcting its terminal basis.

Proposition 4.6 (No automorphism-independent constant). *There is no constant C such that every elementary AC3 move transports through every $\phi \in \text{Aut}(F_2)$ using at most C source-convention AC3 moves via the construction above.*

Proof. Use $\phi_k(x) = xy^k, \phi_k(y) = y$. It is an automorphism with inverse $x \mapsto xy^{-k}, y \mapsto y$. A conjugation by x transports to a conjugation by xy^k , whose freely reduced length is $k+1$ for $k \geq 0$. The constructive source-convention expansion therefore has $k+1$ AC3 moves, which is unbounded as $k \rightarrow \infty$. $\square$

Remark 4.7. Proposition 4.6 concerns a uniform bound over all automorphisms. For each explicitly given ϕ , the bound $\Lambda(\phi)$ is finite and directly computable. This effectiveness is fundamentally different from the unbounded normal-closure parameter in Lemma 11 of the stable-AC discussion.

4.2 From the transformed endpoint back to (x, y)

If an original path ends at (x, y) , its transported path ends at

$$(\phi(x), \phi(y)),$$

not necessarily at (x, y) . Because ϕ is an automorphism, this pair is a free basis.

Theorem 4.8 (Nielsen generation in rank two). *The group $\text{Aut}(F_2)$ is generated by the elementary basis operations*

$$(u, v) \mapsto (v, u), \quad (u, v) \mapsto (u^{-1}, v), \quad (u, v) \mapsto (uv, v),$$

together with the corresponding operations on the second entry.

This is the classical Nielsen generation theorem. On a relator pair, its three displayed generators are respectively relator-order bookkeeping, one AC2, and one AC1. Hence a decomposition of ϕ^{-1} into Nielsen generators gives an explicit source-convention AC sequence from $(\phi(x), \phi(y))$ to (x, y) . This is stronger than merely observing that the terminal pair generates F_2 .

Let $C_{\text{basis}}(\phi)$ denote the number of source-convention AC steps obtained from one chosen, explicitly tracked decomposition of ϕ^{-1} as in Theorem 4.8; swap steps cost zero. Combining it with Corollary 4.4 gives

$$N_{\text{total}} \leq N_1 + N_2 + \Lambda(\phi)N_3 + C_{\text{basis}}(\phi).$$

No claim is made that this is minimal.

The current project certificates perform an untracked Boolean Nielsen basis check: they prove that the endpoint is a basis but do not store a decomposition or a numerical value of $C_{\text{basis}}(\phi)$. Obtaining a numerical endpoint bound therefore requires separately logging a Nielsen reduction. If that reduction uses the common replacements below, they expand constructively as follows:

replacement of a	ab	ab^{-1}	$b^{-1}a$	ba
source-convention cost	1	3	3	5.

For example, $b^{-1}a = (a^{-1}b)^{-1}$ costs AC2, AC1, AC2; the other three bounds follow by the same inversion-and-restoration constructions. Any terminal signs require at most two additional AC2 moves.

Example 4.9 (Endpoint cost in the source convention). For $(\phi(x), \phi(y)) = (xy, y)$, the tempting operation $xy \mapsto xy y^{-1} = x$ is not one source-convention AC1 move. It is the three-move sequence

$$(xy, y) \xrightarrow{\text{AC2}} (xy, Y) \xrightarrow{\text{AC1}} (xyY, Y) = (x, Y) \xrightarrow{\text{AC2}} (x, y).$$

Thus this chosen source-convention endpoint correction has cost 3. Under a macro convention allowing multiplication by r_2^{-1} , it would be encoded as one macro-move.

Theorem 4.10 (Automorphism invariance of AC-triviality). *For every balanced two-generator presentation P and every $\phi \in \text{Aut}(F_2)$,*

$$P \text{ is AC-trivial} \iff \phi(P) \text{ is AC-trivial}.$$

Proof. Assume P is AC-trivial. Transport its finite AC path to obtain

$$\phi(P) \xrightarrow{\text{AC}^*} (\phi(x), \phi(y)).$$

Choose a finite Nielsen-generator decomposition of ϕ^{-1} ; it gives a finite AC sequence from the terminal basis to (x, y) . Hence $\phi(P)$ is AC-trivial. Conversely, apply the same argument to ϕ^{-1} . $\square$

$$\begin{array}{ccc}
P & \xrightarrow{\text{AC}^*} & (x, y) \\
\downarrow \phi & & \downarrow \phi \\
\phi(P) & \xrightarrow{\text{AC}^*} & (\phi(x), \phi(y)) \xrightarrow{\text{AC}^*} (x, y).
\end{array}$$

Warning 4.11 (What the theorem does not say). It does not prove $P \sim_{\text{AC}} \phi(P)$ for arbitrary P . An attempted chain

$$P \sim_{\text{AC}} (x, y) \sim_{\text{AC}} (\phi(x), \phi(y)) \sim_{\text{AC}} \phi(P)$$

already assumes the unknown first link. The theorem says that P and $\phi(P)$ have the same truth value for AC-triviality; it does not say that they lie in the same ordinary AC component.

Exercises

Exercise 4.1. For $\phi(x) = xy, \phi(y) = y$, transport the AC3 move

$$r_1 \mapsto xr_1X$$

step by step, using only generator conjugations.

Exercise 4.2. An original path has 10 AC1 moves, 4 AC2 moves, 7 conjugations by x , and 3 conjugations by y . For the automorphism $\phi(x) = xyY, \phi(y) = Yx$, compute the direct bound from Corollary 4.4, after freely reducing the generator images.

Exercise 4.3. Prove that one source-convention AC1 and one AC2 always transport to one move of the same type, even if substantial free cancellation occurs in the displayed words.

Exercise 4.4. Explain precisely why Theorem 4.1 should use $\xrightarrow{\text{AC}^*}$ rather than a single arrow under the source-paper convention, but may use a single arrow for AC3 under an arbitrary-word conjugation convention.

Exercise 4.5. Find the logical error in the statement: “Since P and $\phi(P)$ present isomorphic trivial groups, they are AC-equivalent.”

5 Canonicalization and the eight signed relabelings

5.1 Canonicalization

The search stores a canonical representative of a relator pair. For each relator it:

1. freely and cyclically reduces it;
2. considers the relator and its inverse;
3. selects the lexicographically least cyclic rotation;
4. sorts the two canonical relators by length and lexical order.

These operations do not invent an equivalence.

- Free reduction preserves the element of F_2 .

- Removing an outer inverse pair $awa^{-1} \mapsto w$ is one source-convention AC3 move, followed by free reduction.
- Whole-relator inversion is AC2.
- If $r = uv$, the rotation $uv \mapsto vu$ is conjugation:

$$vu = u^{-1}(uv)u.$$

Under generator-only AC3 it may require several elementary conjugations.

- Relator order is bookkeeping.

Canonicalization is therefore safe for AC-triviality, although it should not be assigned zero elementary AC cost when one is explicitly counting source-convention moves.

5.2 The eight signed permutations

A signed permutation sends x to one of x, X, y, Y and y to a signed version of the other underlying generator. There are

$$2! 2^2 = 8$$

such maps:

$$\begin{array}{cccc} (x, y), & (X, y), & (x, Y), & (X, Y), \\ (y, x), & (Y, x), & (y, X), & (Y, X). \end{array}$$

Each is an automorphism whose inverse is another signed permutation.

For every input pair P , the relabeling key is

$$K_{\text{signed}}(P) = \min_{\sigma \in \Sigma_8} \text{canon}(\sigma(P)).$$

Equal keys mean the two presentations differ by a signed change of free basis plus canonical bookkeeping. By Theorem 4.10, they have the same AC-triviality status.

Signed permutations are particularly well behaved computationally: they preserve word length and commute with inversion and cyclic rotation. Consequently their child sets under the encoded S-move agree after relabeling. General automorphisms such as $x \mapsto xy$ do not have this property because they change word lengths and rotation positions.

5.3 Worked example: 19_45 and 19_50

The presentations have canonical relator pairs

$$\begin{aligned} 19_45 &= (YYYxxyyX, YYYXyxxYxx), \\ 19_50 &= (YYYxyyXX, YYYXXYXXyx). \end{aligned}$$

Use the signed automorphism

$$\sigma(x) = x, \quad \sigma(y) = Y.$$

For the first relator:

$$YYYxxyyX \xrightarrow{\sigma} yyyxxYYX \xrightarrow{\text{invert}} xyyXXYYY \xrightarrow{\text{right rotate 3}} YYYxyyXX.$$

For the second:

$$YYYYYxYxxYxx \xrightarrow{\sigma} yyyYXYxyxx \xrightarrow{\text{invert}} XXYXXyxYYYY \xrightarrow{\text{right rotate 4}} YYYYYXXYXXyx.$$

Thus

$$\text{canon}(\sigma(19_45)) = 19_50.$$

No search is needed. The substitution is the mathematical change of variables; inversion and rotation explain why the canonical strings do not match immediately.

5.4 Counts

The 261 exact input pairs remain 261 after canonicalization: there are no root collisions. Quotienting by the eight signed permutations yields

$$261 \longrightarrow 171.$$

This identifies 90 redundant rows.

Exercises

Exercise 5.1. List the eight signed permutations and write the inverse of each one.

Exercise 5.2. Repeat the first-relator computation for $19_45 \rightarrow 19_50$ without looking at the worked derivation. Explain why simply changing the case of y does not immediately produce the stored target.

Exercise 5.3. Prove that signed permutations commute with free reduction and cyclic rotation. Identify the step that fails for $\phi(x) = xy, \phi(y) = y$.

Exercise 5.4. If a signed-relabeling class has eight members and one representative is proved AC-trivial in 200 moves, what can be concluded about the other seven? What cannot be concluded about their shortest path lengths?

6 Full Aut(F_2) and Whitehead canonicalization

The eight signed permutations form only a small subgroup of Aut(F_2). Length-changing Nielsen transformations, such as

$$x \mapsto xy, \quad y \mapsto y,$$

can identify presentations missed by signed relabeling.

6.1 The twenty elementary Whitehead generators

The implementation does not invent thousands of ad hoc maps. `experiments/equivalence_classes/lib/autcanon.py` always tries the same finite list of **20** elementary Whitehead automorphisms. Peak reduction and the length-preserving level-set search *compose* these maps; a certificate may therefore record a composite such as $x \mapsto xxxxxxxy, y \mapsto X$, which is not itself one of the twenty generators.

Eight first-kind maps (signed permutations).

$$\begin{array}{cccc} (x, y), & (x, Y), & (X, y), & (X, Y), \\ (y, x), & (y, X), & (Y, x), & (Y, X). \end{array}$$

Twelve second-kind maps (length-changing).

$$\begin{array}{lll} x \mapsto x, y \mapsto yx & x \mapsto x, y \mapsto Xy & x \mapsto x, y \mapsto Xyx \\ x \mapsto x, y \mapsto yX & x \mapsto x, y \mapsto xy & x \mapsto x, y \mapsto xyX \\ x \mapsto xy, y \mapsto y & x \mapsto Yx, y \mapsto y & x \mapsto Yxy, y \mapsto y \\ x \mapsto xY, y \mapsto y & x \mapsto yx, y \mapsto y & x \mapsto yxY, y \mapsto y. \end{array}$$

Only the twelve second-kind maps are used in the length-descent loop; the eight signed permutations already preserve total length.

6.2 The classical theorem used

Theorem 6.1 (Whitehead peak reduction, specialized statement). *For a finite tuple of cyclic words in a finitely generated free group:*

1. *if the tuple is not of minimum total length in its automorphism orbit, some Whitehead automorphism strictly decreases its total length;*
2. *all minimum-length representatives in the same orbit are connected through length-preserving Whitehead moves.*

This classical theorem is cited rather than reproved. Its hypotheses apply to the object used here: an unordered pair of cyclic relators, with each relator considered up to cyclic rotation and inversion. The function canon chooses a word-pair representative of this object.

The project converts Theorem 6.1 into a complete orbit key:

1. repeatedly apply a length-decreasing Whitehead automorphism;
2. once minimum total length is reached, explore the entire length-preserving minimum level set;
3. choose the lexicographically least canonical pair in that set.

Proposition 6.2. *For canonical relator pairs P, Q , their Whitehead keys satisfy*

$$K(P) = K(Q) \iff \exists \phi \in \text{Aut}(F_2) \text{ such that } \text{canon}(\phi(P)) = \text{canon}(Q).$$

Proof. An orbit of unordered cyclic-relator pairs modulo inversion has one minimum-level set. By Theorem 6.1, every descent reaches its minimum length and every minimum representative lies in the connected level set explored in step 2. Hence the lexicographic minimum is independent of the starting canonical pair. Equal keys therefore imply the displayed canonicalized automorphism relation, while that relation necessarily produces the same key. $\square$

Peak reduction alone is not a canonical key: distinct descent choices can reach different minimum representatives. On the 261 inputs, using only the peak-reduced endpoint would incorrectly produce 259 classes rather than 168. Exploring the complete minimum level set is essential.

6.3 Independent checks

The exact partition was computed independently by:

- `experiments/analysis/whitehead.py`, and
- `experiments/equivalence_classes/lib/autcanon.py`.

They agree element for element. The second implementation records a witness ϕ satisfying

$$\text{canon}(\phi(P)) = \text{reported orbit representative}.$$

The certificate verifier then proves independently, by Nielsen reduction, that every recorded pair $(\phi(x), \phi(y))$ is a basis. Thus a bug that produced a non-invertible substitution would be rejected.

6.4 Empirical Aut witnesses on the 93 pure change-of-variable edges

The shipped proof book `results/equivalence_classes/certificates.json` contains 137 verified edges in total:

$$93 \text{ cv/aut} + 44 \text{ ac/aca} = 137.$$

The 93 edges are **pure automorphisms**: a single substitution ψ with $\text{canon}(\psi(A)) = \text{canon}(B)$ and *no* Andrews–Curtis move. The 44 remaining edges are the ones that use Definition 2.1 S-moves (ACA search); they are not counted in the table below.

Among those 93 pure Aut edges, the witnessing substitution ψ has the following empirical frequencies:

Count	Map ψ	Notes
82	$x \mapsto x, y \mapsto Y$	dominates; signed
3	$x \mapsto Y, y \mapsto x$	signed
2	$x \mapsto y, y \mapsto x$	signed
1	$x \mapsto X, y \mapsto y$	signed
1	$x \mapsto xy, y \mapsto y$	second-kind / composite
1	$x \mapsto xY, y \mapsto y$	second-kind / composite
1	$x \mapsto XY, y \mapsto y$	longer word
1	$x \mapsto xY, y \mapsto Y$	mixed
1	$x \mapsto yx, y \mapsto Y$	mixed

Thus 88 of the 93 pure Aut merges are single-letter (signed-permutation style), and only 5 need a longer generator image. The textbook example $x \mapsto xy, y \mapsto y$ appears, but only once as a direct cv witness; signed $y \mapsto Y$ accounts for almost all Aut collapsing of the 261 inputs.

6.5 Worked example: 19_37 and 17_41

Take

$$\begin{aligned} 19_37 &= (\text{YYXXyx}, \text{YXyXyXyxxYxYx}), \\ 17_41 &= (\text{YYXYXyx}, \text{YXYxxxYXXX}). \end{aligned}$$

Apply $\phi(x) = xy, \phi(y) = y$, proved invertible in Example 3.2. The proof book records:

$$YYXXyx \xrightarrow{\phi} YYYXXYxy \xrightarrow{\text{reduce}} YYYXXyx,$$

and

$$\begin{aligned} w_0 &= YXyXyXyxxYxYx, \\ w_1 &= YYXXXyxyxxxy \quad (\text{apply } \phi), \\ w_2 &= YXXXyxyxxx \quad (\text{freely reduce}), \\ w_3 &= XXXYXYxxxy \quad (\text{invert}), \\ w_4 &= YXYxxxyXXX \quad (\text{right rotate by 7}). \end{aligned}$$

Therefore

$$\text{canon}(\phi(19_37)) = 17_41.$$

This is a full-Aut(F_2) identification not restricted to one-letter signed relabeling.

6.6 Exact count

Whitehead canonicalization gives

$$171 \longrightarrow 168.$$

Unlike the later search-dependent numbers, 168 is exact with respect to change of variables: no further Aut(F_2) orbit mergers exist among the 261 inputs.

Exercises

Exercise 6.1. Explain why the map $x \mapsto xy, y \mapsto y$ can change which numerical rotations exist even though it preserves AC-triviality.

Exercise 6.2. Why does a witness satisfying $\text{canon}(\phi(P)) = \text{canon}(Q)$ fail to prove anything unless ϕ is verified to be an automorphism?

Exercise 6.3. Give a concise proof that equality of complete Whitehead canonical keys is an equivalence relation. Which part depends on Theorem 6.1?

Exercise 6.4. Explain why agreement of two independently written implementations is stronger evidence than running one implementation twice, but is not itself a replacement for the mathematical theorem.

7 Definition 2.1 S-moves as elementary AC sequences

The equivalence search does not expand elementary AC1–AC3 one at a time. It uses the solver's substitution or S-move encoding:

$$r_i \longmapsto \text{rot}_{k_1}(r_i) \text{rot}_{k_2}(r_j^\varepsilon), \quad i \neq j, \quad \varepsilon \in \{+1, -1\},$$

followed by reduction and canonicalization. The other relator must remain unchanged in the mathematical presentation.

7.1 Rotations are conjugates

Let $r = uv$, and let the rotation move the suffix v to the front:

$$\text{rot}(r) = vu.$$

There are two useful conjugation identities:

$$vu = u^{-1}(uv)u = v(uv)v^{-1}.$$

Thus one may choose either conjugator u^{-1} or v . Under generator-only AC3, a conjugator $c = a_1 \cdots a_m$ expands into m generator conjugations. Define $C(r, k)$ to be the length of the chosen freely reduced conjugator realizing $\text{rot}_k(r)$. A convenient bound is

$$C(r, k) \leq \min\{k, |r| - k\}$$

for a right rotation by k , before any further cancellation.

7.2 A constructive elementary realization

Let

$$R_i = c_1 r_i c_1^{-1} = \text{rot}_{k_1}(r_i), \quad R_j = c_2 r_j^\varepsilon c_2^{-1} = \text{rot}_{k_2}(r_j^\varepsilon),$$

and put $m_1 = |c_1|$, $m_2 = |c_2|$. The following source-convention sequence replaces r_i by $R_i R_j$ and restores the other relator to r_j :

1. If $\varepsilon = -1$, invert r_j using AC2.
2. Conjugate r_i by c_1 , using m_1 generator AC3 moves.
3. Conjugate the current r_j^ε by c_2 , using m_2 generator AC3 moves.
4. Apply one AC1 to replace the current target by its product with the current other relator.
5. Conjugate the other relator by c_2^{-1} , using m_2 AC3 moves, restoring it to r_j^ε .
6. If $\varepsilon = -1$, invert the other relator again, restoring r_j .

Proposition 7.1 (Source-convention S-move bound). *Before final canonicalization, the construction above uses at most*

$$\boxed{1 + m_1 + 2m_2 + 2 \mathbf{1}_{\{\varepsilon = -1\}}}$$

elementary source-convention moves.

Proof. There is one AC1, $m_1 + 2m_2$ generator AC3 moves, and two AC2 moves exactly when $\varepsilon = -1$. Inspection of the six stages proves that the target is $R_i R_j$ and the other relator is restored. $\square$

This is a constructive upper bound, not a shortest-path formula. Other realizations may be shorter. If arbitrary-word conjugation is counted as one AC3 macro, the corresponding construction uses at most one AC1, three arbitrary-word conjugations (target, other, restore other), and two inversions in the negative case. This explains why different papers quote different expansion counts.

7.3 Canonicalization cost

The encoded child is then freely/cyclically reduced, possibly inverted, rotated, and swapped. To convert the raw child to that exact stored pair under the source convention, add:

- zero for free reduction and relator sorting;
- one AC2 for each whole-relator inversion selected;
- one generator AC3 per letter in each chosen cyclic-reduction or rotation conjugator.

Therefore the exact certificate supplies enough information to compute a finite elementary upper bound, but the encoded S-move count itself is not the elementary AC count.

7.4 Transporting an S-move through ϕ

One may first expand the S-move into elementary operations and then apply Proposition 4.3. Equivalently, replace each conjugator c by $\phi(c)$. Since

$$|\phi(c)| \leq \Lambda(\phi) |c|,$$

the pre-canonicalization transported construction has the bound

$$1 + |\phi(c_1)| + 2|\phi(c_2)| + 2\mathbf{1}_{\{\varepsilon=-1\}}$$

and hence

$$1 + \Lambda(\phi)(m_1 + 2m_2) + 2\mathbf{1}_{\{\varepsilon=-1\}}.$$

Free cancellation can only lower these lengths.

Warning 7.2 (Why child sets do not commute with general automorphisms). The raw algebraic identity transports correctly, but the solver enumerates rotations by numerical positions in the current words. A length-changing automorphism changes those positions and may introduce cancellation. Therefore, for general ϕ ,

$$\text{children}(\phi(P)) \neq \phi(\text{children}(P))$$

as enumerated canonical sets. This does not threaten soundness: every edge found is still legal. It causes incompleteness because some quotient edges may be missed when only one representative of an Aut-orbit is expanded.

Exercises

Exercise 7.1. For $r = abcdef$, express the right rotation by 2, $efabcd$, as conjugation in two different ways. Compare the two conjugator lengths.

Exercise 7.2. Use Proposition 7.1 when $m_1 = 2, m_2 = 3, \varepsilon = -1$. Give the number of AC1, AC2, and AC3 moves separately.

Exercise 7.3. Write all intermediate relator pairs in the six-stage construction when $c_1 = x$, $c_2 = Y$, and $\varepsilon = -1$. Do not simplify away the restoration steps.

Exercise 7.4. Let $\phi(x) = xy, \phi(y) = y$. If $c_1 = Xy$ and $c_2 = xY$, calculate $\phi(c_1), \phi(c_2)$, freely reduce them, and evaluate the sharper transported S-move bound for $\varepsilon = +1$.

8 ACA: the equivalence relation used by the search

Definition 8.1. Write $P \sim_{\text{ACA}} Q$ when P and Q are connected by a finite sequence whose steps are either:

1. ordinary AC moves, or
2. changes of variables by automorphisms in $\text{Aut}(F_2)$.

An automorphism step is not declared to be an AC step. Instead, the important Boolean property

$$\mathcal{T}(P) = \begin{cases} 1, & P \text{ is AC-trivial,} \\ 0, & P \text{ is not AC-trivial} \end{cases}$$

is invariant under each type of ACA step.

Proposition 8.2 (ACA preserves the problem). *If $P \sim_{\text{ACA}} Q$, then*

$$P \text{ is AC-trivial} \iff Q \text{ is AC-trivial.}$$

Proof. An AC move preserves membership in the AC component of (x, y) , because it is reversible by a finite AC sequence. An automorphism preserves AC-triviality by Theorem 4.10. Equality of the truth value propagates along a finite chain. $\square$

Warning 8.3. “Same ACA problem” is weaker than “joined by a raw AC path.” The search therefore reports components of a finite verified ACA-edge subgraph, not proven ordinary AC-equivalence classes and not an exact partition into full ACA classes.

8.1 The quotient search graph

The search uses:

- **Node:** one complete $\text{Aut}(F_2)$ -orbit, keyed by its Whitehead canonical representative.
- **Edge:** apply one encoded S-move to the chosen Aut-minimal representative, then map the child to its Whitehead key.
- **Priority:** the Aut-minimal total relator length.

Using Aut-minimal length matters. A raw word pair of total length 34 may be one change of variables away from length 16. A raw length-ordered search must enumerate a large “hump” that the quotient representation collapses.

8.2 Soundness and incompleteness

Theorem 8.4 (Soundness of every reported merge). *If two sources meet in the quotient search, they have the same AC-triviality status.*

Proof. Each path step consists of a legal S-move, which Proposition 7.1 expands into elementary AC moves, followed by a verified automorphism and canonical bookkeeping. Each part preserves $\mathcal{T}$. If paths from P and Q end in the same Aut-orbit, transitivity gives $\mathcal{T}(P) = \mathcal{T}(Q)$. $\square$

The search is incomplete for three reasons:

1. it expands only one representative from each Aut-orbit;
2. general automorphisms do not preserve the enumerated child set;
3. length, state, time, and memory caps remove possible states.

All three mechanisms remove edges; none can create a false edge. Hence:

Every merge found is a proof. No merge found proves nothing.

Exercises

Exercise 8.1. Construct a three-step ACA chain containing one automorphism step and two AC steps. Explain why it proves equality of AC-triviality status without proving that the endpoints are directly adjacent in the AC graph.

Exercise 8.2. Why does expanding only one representative of an Aut-orbit preserve soundness but lose completeness?

Exercise 8.3. Suppose a search cap removes the only path connecting two reported components. Does this create too many components or too few? Why is this the safe direction for choosing a workload?

Exercise 8.4. State the precise difference between a node of the raw AC graph and a node of the ACA quotient graph.

9 The complete reduction: 261 to at most 124

Identification applied	Count	Status
Exact ordered relator pair	261	exact
Canonical rotation/inversion/swap	261	exact; no collisions
Eight signed generator relabelings	171	exact for this subgroup
Full Aut(F_2), Whitehead canonical key	168	exact Aut-orbit partition
ACA search, Aut-minimal ceiling 28	126	sound upper bound
Probe at ceiling 34	125	sound upper bound
Overnight ladder, ceilings 30–36	124	sound upper bound

9.1 From 168 to 126

The canonical cap-28 sweep seeded all 261 roots and the trivial presentation. Initial Aut coincidences supplied 93 change-of-variable forest edges, reducing 261 to 168. The ACA search supplied 42 further verified edges, reducing

$$168 \longrightarrow 126.$$

The resulting 135 edges form a spanning forest on the 261 input vertices:

$$261 - 135 = 126.$$

This historical 126-component forest is recorded in `results/equivalence_classes/sweep_seam_28_250.json`. The current proof book and certificates were regenerated after adding the two later edges and contain 124 components.

9.2 Why 126 became 125

The cap-28 search could not reach an Aut-minimal meeting state of total length 30, regardless of node budget. A probe with ceiling 34 found

$$21_3 \sim_{\text{ACA}} 21_29,$$

joining two singleton components and giving 125. The paths had lengths one and three encoded S-moves to a common Aut-orbit.

9.3 Why 125 became 124

Five overnight arms at ceilings 30, 32, 32 (deterministic rerun), 34, and 36 all found the additional merge

$$21_7 \sim_{\text{ACA}} 21_28.$$

Again the two sides reached a common Aut-orbit in one and three encoded S-moves. This joins another pair of singleton components:

$$126 - 2 = 124.$$

The five arms explored roughly 700,000 nodes in total and found no further merger, but every arm ended because of a state or memory resource cap, not because its queue was exhausted.

9.4 The coverage theorem

Theorem 9.1 (One representative per certified component suffices). *Let $\mathcal{P}$ be the 261 input presentations, and let the verified ACA edges have 124 connected components. Choose one representative R_C from each component C . If every R_C is AC-trivial, then every presentation in $\mathcal{P}$ is AC-trivial.*

Proof. For every $P \in \mathcal{P}$, there is exactly one certified component C containing it and a finite verified edge path from P to R_C . By Proposition 8.2,

$$P \text{ is AC-trivial} \iff R_C \text{ is AC-trivial.}$$

The assumed triviality of all 124 representatives therefore implies the triviality of all 261 inputs. $\square$

Corollary 9.2 (Safe-direction error). *If the incomplete search missed additional valid mergers, running 124 representatives performs redundant work but does not omit any of the 261 inputs.*

Proof. A missed edge can only merge two currently separate certified components, so the true ACA partition can be coarser and have fewer components. It cannot split a verified component. Every original vertex is already in one selected component. $\square$

9.5 How to obtain a 124-representative workload

The historical probe-seed manifest `results/equivalence_classes/classes_126_from_greedy_1000000_261_mrl48.jsonl` still contains 126 rows and is intentionally retained because the probe certificates name its class IDs. Starting from that historical manifest, merge:

$$\begin{array}{l} C123 (21_29) \text{ with } C124 (21_3), \\ C122 (21_28) \text{ with } C125 (21_7). \end{array}$$

Keep either representative from each pair. All other canonical components remain unchanged, producing 124 representatives. The current production deliverable has already performed these unions:

`results/equivalence_classes/classes_sweep_seam_28_250_plus_overnight.csv`.

It contains 124 rows and supersedes the historical 126-row workload.

9.6 What 124 does and does not mean

- It means there are *at most* 124 distinct ACA problems among these 261 inputs.
- It does not mean that the 124 certified components are mutually ACA-inequivalent; more search may merge them.
- It does not mean there are 124 ordinary AC-classes.
- It does not prove any input AC-trivial. The reported solved set is empty.
- It proves a conditional workload statement: 124 successful trivializations would settle all 261.

Exercises

Exercise 9.1. Verify the component arithmetic

$$261 - 93 - 42 - 2 = 124.$$

Explain what each subtraction represents mathematically.

Exercise 9.2. Suppose a future verified run finds five additional edges, but two lie inside already connected components. By how much can the component count decrease?

Exercise 9.3. Give a formal counterexample to the statement: “Because the search returned 124 components, it proved there are at least 124 distinct problems.”

Exercise 9.4. If only 20 of the selected 124 representatives are trivialized, what exactly has been proved about the 261 inputs?

Exercise 9.5. Explain why five runs agreeing on 124 strengthens empirical confidence but does not turn the upper bound into an exact count.

10 Certificates: why the mergers are proofs

The search output is not trusted merely because it prints a class count. The verification layers are deliberately separated.

10.1 Two edge types

The current proof book has exactly 137 edges. They split as follows.

cv/aut — 93 edges. **Pure Aut / change of variables only.** A recorded automorphism ψ must satisfy

$$\text{canon}(\psi(A)) = \text{canon}(B).$$

No Definition 2.1 S-move appears. These 93 forest edges are exactly what collapses $261 \rightarrow 168$ Aut-orbits. Empirically, 82 of them use $\psi: x \mapsto x, y \mapsto Y$; see the frequency table in Section 6.

ac/aca — 44 edges. **ACA edges: AC/S-moves are required.** Both roots have replayable Definition 2.1 paths, optionally interleaved with verified automorphisms, ending in the same Aut-orbit. The historical cap-28 sweep found 42 such edges ($168 \rightarrow 126$); the two later probe mergers ($21_3 \sim 21_29$ and $21_7 \sim 21_28$) bring the current book to 44. Of these 44, only 6 are tagged pure AC paths (every recorded Aut witness on the path is the identity after Aut-minimal alignment); the rest mix nontrivial ϕ with S-moves.

$$\underbrace{93}_{\text{Aut only}} + \underbrace{44}_{\text{ACA with S-moves}} = 137 \quad \text{and} \quad 261 - 137 = 124.$$

An **aca** edge generally does not satisfy $\text{canon}(\psi(A)) = \text{canon}(B)$ for one direct substitution; AC moves occur between the endpoints. Applying the pure-change test to an **aca** edge would reject a correct certificate.

10.2 Independent verification

The verifier performs the following checks.

1. Re-read each source presentation from the original CSV, binding both its row identity and its relator words.
2. Replay each Definition 2.1 move by direct word operations.
3. Check every recorded ϕ by independent Nielsen basis reduction.
4. Canonicalize independently of the pure-Python search routine.
5. Require both sides of an ACA merge to end in the same recorded Aut-orbit.
6. Check the exponent-sum determinant invariant.
7. Rebuild the entire partition using only edges that passed all previous checks.

The last step is essential. Checking each listed edge cannot detect a separate union-find bug that assigns an unconnected vertex to a component. Rebuilding the transitive closure from verified edges detects such an over-merge.

The production verifier

`experiments/equivalence_classes/verify/verify_proofs.py`.

re-proves the current 137-edge, 124-component proof book from `results/equivalence_classes/certificates.json` and the raw input CSV. The two mergers that historically changed 126 to 124 are also checked directly in their original probe files by

`experiments/equivalence_classes/verify/verify_probe_merges.py`.

The latter reconstructs roots from the historical 126 manifest rather than trusting the probe files' own root blocks.

10.3 Mutation testing

The checker is tested by deliberately corrupting certificates: replacing an automorphism with a non-basis endomorphism, changing a rotation number, swapping words between presentation IDs, corrupting a concatenation, or forcing an unsupported over-merge. Each mutation must cause verification to fail. This tests not only the mathematical certificate but also whether the checker actually enforces the advertised conditions.

10.4 Invariant cross-check

For a pair $P = (r_1, r_2)$, form the exponent-sum matrix

$$M(P) = \begin{pmatrix} \exp_x(r_1) & \exp_y(r_1) \\ \exp_x(r_2) & \exp_y(r_2) \end{pmatrix}.$$

Elementary AC1 and AC2 perform unimodular row operations; AC3 does not change exponent sums. A change of free basis performs a unimodular column operation. Therefore

$$|\det M(P)|$$

is ACA-invariant. It equals 1 for every input, so it cannot separate the remaining groups, but a mismatch across a claimed merge would expose an error.

Exercises

Exercise 10.1. Why is checking every edge insufficient to verify the reported component count?

Exercise 10.2. Show directly how AC1, AC2, and AC3 act on the exponent-sum matrix and prove that $|\det M|$ is unchanged.

Exercise 10.3. Explain why the fact that $|\det M| = 1$ for all 261 is useful as a bug detector but useless as a discriminator among these inputs.

Exercise 10.4. Design one certificate mutation not listed above and state the verifier condition that should reject it.

11 Presentation-ready synthesis

11.1 A concise theorem chain

1. A balanced presentation $\langle x, y \mid r_1, r_2 \rangle$ is the quotient $F_2 / \langle\langle r_1, r_2 \rangle\rangle$.
2. An automorphism $\phi \in \text{Aut}(F_2)$ is a reversible change of free basis and induces an isomorphism of the presented quotient groups.
3. Every elementary AC move transports through ϕ to a finite source-convention AC sequence. AC1 and AC2 cost one; AC3 by g costs at most $|\phi(g)|$.
4. Hence P is AC-trivial if and only if $\phi(P)$ is AC-trivial.
5. Therefore each AC or Aut edge preserves the question “is this presentation AC-trivial?”
6. The finite verified ACA-edge subgraph on 261 inputs has exactly 124 components, each contained in a true ACA class.
7. Consequently one representative per component is a sufficient coverage set; proving all 124 representatives AC-trivial would prove all 261 AC-trivial.

11.2 Move-bound summary

Object	Constructive source-convention bound
Transported AC1	1
Transported AC2	1
Transported AC3 by g	$ \phi(g) \leq \Lambda(\phi)$
Transported path	$N_1 + N_2 + \Lambda(\phi)N_3$, before basis correction
Terminal basis correction	$C_{\text{basis}}(\phi)$, from a separately logged Nielsen reduction; not stored by the current certificates
One raw S-move before canonicalization	$1 + m_1 + 2m_2 + 2\mathbf{1}_{\{\varepsilon=-1\}}$
Transported raw S-move	$1 + \Lambda(\phi)(m_1 + 2m_2) + 2\mathbf{1}_{\{\varepsilon=-1\}}$

11.3 Claims to make

- “The 261 inputs represent at most 124 distinct ACA problems.”
- “The 168 full-Aut(F_2) count is exact.”
- “The 124 count is the transitive closure of certified mergers and is still an upper bound.”
- “One representative per certified component covers all 261.”
- “Zero of the 261 were trivialized by this analysis.”

11.4 Claims not to make

- “There are exactly 124 AC-classes.”

- “The 124 certified components are proven ACA-inequivalent.”
- “ P is always AC-equivalent to $\phi(P)$.”
- “An automorphism preserves the same numerical rotation or the same elementary move count.”
- “The solver’s r_j^{-1} multiplication is one elementary AC1 move under the source convention.”
- “The equivalence-class computation solved any presentation.”

11.5 Oral rehearsal questions

Exercise 11.1. Give a two-minute explanation of why $\phi(x) = xy, \phi(y) = y$ is an automorphism and why it preserves AC-triviality.

Exercise 11.2. Give a one-minute explanation of why 124 is sufficient but not exact.

Exercise 11.3. An audience member asks: “If ACA-equivalence is not AC-equivalence, why are you allowed to discard 137 runs?” Give a rigorous response.

Exercise 11.4. An audience member asks: “Did you prove 124 presentations trivial?” Answer without weakening or overstating the result.

12 Complete answer key and learning diagnostics

The answers deliberately show intermediate reasoning. When an exercise can be solved in several ways, the answer gives one auditable construction rather than claiming minimality.

Answers for Section 1: AC conventions

Answer to Exercise 1.1

Invert the second relator, multiply, and restore it:

$$(r_1 r_2, r_2) \xrightarrow{\text{AC2}} (r_1 r_2, r_2^{-1}) \xrightarrow{\text{AC1}} (r_1 r_2 r_2^{-1}, r_2^{-1}) = (r_1, r_2^{-1}) \xrightarrow{\text{AC2}} (r_1, r_2).$$

This also proves that the directed-looking positive AC1 operation is reversible by a finite sequence, which is enough for AC-equivalence.

Diagnostic: If the middle step looked like multiplication by an inverse, notice that the current second relator is already r_2^{-1} . AC1 is still positive multiplication by the current relator.

Answer to Exercise 1.2

If AC3 permits $r \mapsto wrw^{-1}$ for arbitrary w , then any cyclic rotation is one AC3 macro-move. In the source-paper convention, AC3 permits only one generator or inverse. Conjugation by a reduced word $w = a_1 \cdots a_m$ then expands into m elementary AC3 moves. Thus “one rotation” may mean one encoded operation or several elementary source-paper moves.

Diagnostic: Whenever a move count is quoted, first write the permitted conjugator beside AC3.

Answer to Exercise 1.3

- $r_1 \mapsto r_1^{-1}$ is one elementary AC2.
- $r_1 \mapsto r_1 r_2^{-1}$, while restoring r_2 , is a three-move macro: AC2, AC1, AC2.
- $r_1 \mapsto wr_1 w^{-1}$ is elementary only when the freely reduced w is one generator or inverse. Otherwise it is an arbitrary-word macro that expands into $|w|$ generator AC3 moves.
- $(r_1, r_2) \mapsto (r_2, r_1)$ is bookkeeping because relator order is not part of the presentation.

Diagnostic: The operation name alone is insufficient; check both the allowed operand and what must remain unchanged.

Answers for Section 2: free groups and presentations

Answer to Exercise 2.1

Reduce from left to right:

$$xyYXxXY \longrightarrow xXxXY \longrightarrow xXY \longrightarrow Y.$$

More explicitly, yY , then xX , then the remaining xX cancel. The reduced word is Y , whose inverse is y .

Diagnostic: If a different answer was obtained, use a stack: cancel only adjacent inverse letters and re-check adjacency after each cancellation.

Answer to Exercise 2.2

The relation $y = 1$ and the relation $xy = 1$ imply $x = 1$, so the presented group is trivial. Under the source convention:

$$(xy, y) \xrightarrow{\text{AC2}} (xy, Y) \xrightarrow{\text{AC1}} (xyY, Y) = (x, Y) \xrightarrow{\text{AC2}} (x, y).$$

Thus the presentation is not merely group-trivial; this explicit sequence proves it AC-trivial.

Diagnostic: Separating the quotient argument from the AC sequence is important: the first proves the group is trivial, while the second proves the stronger AC-triviality statement.

Answer to Exercise 2.3

$G_P = 1$ says that r_1, r_2 normally generate F_2 . A normal-closure membership statement permits an unspecified product of conjugates and gives no prescribed sequence reducing the *ordered relator pair* through AC1–AC3. Asserting that such a path always exists is precisely the Andrews–Curtis conjecture.

Diagnostic: Do not replace “normally generates” by “has a known AC derivation.” They are different strengths of information.

Answer to Exercise 2.4

Balancedness means that the number of relators equals the number of generators. The standard AC conjecture compares balanced n -generator, n -relator presentations to the n -generator trivial presentation. The equivalence pipeline fixes $n = 2$ and stores two relator slots. A two-generator, three-relator presentation is not balanced and is neither a vertex in that graph nor one of the 261 inputs.

Diagnostic: Stable AC may temporarily change the number of generators and relators together, but stabilization is not used in the 261-to-124 pipeline.

Answers for Section 3: automorphisms

Answer to Exercise 3.1

Define

$$\psi_k(x) = xy^{-k}, \quad \psi_k(y) = y.$$

Then

$$\psi_k(\phi_k(x)) = \psi_k(xy^k) = xy^{-k}y^k = x, \quad \psi_k(\phi_k(y)) = y.$$

Similarly,

$$\phi_k(\psi_k(x)) = \phi_k(xy^{-k}) = xy^k y^{-k} = x, \quad \phi_k(\psi_k(y)) = y.$$

The two homomorphisms are inverse on the free generators and hence inverse on all of F_2 .

Diagnostic: It is enough to check compositions on x, y only because of the universal property, not because longer words are being ignored.

Answer to Exercise 3.2

The forced letter images are

$$\phi(Y) = Y, \quad \phi(X) = YX, \quad \phi(y) = y, \quad \phi(x) = YX.$$

Therefore

$$\phi(YXyX) = Y YX y YX = YYXyYX.$$

The adjacent yY cancels, leaving

$$\boxed{YYXX}.$$

Diagnostic: For a capital letter, invert the whole image word: reverse it and invert every letter. A simple case swap is wrong for multi-letter images.

Answer to Exercise 3.3

The universal property gives an endomorphism because any two image words define a homomorphism. Let $\exp_x(w)$ denote the exponent sum of x . For every word w ,

$$\exp_x(\eta(w)) = 2 \exp_x(w),$$

which is even. The element $x \in F_2$ has exponent sum 1, so it is not in the image of η . Thus η is not surjective and cannot be an automorphism.

Diagnostic: A determinant or exponent-sum test can disprove invertibility, but passing an abelianized test is not generally enough to prove a free-group endomorphism invertible.

Answer to Exercise 3.4

The transformed relators are (xy, y) . Their normal closure is all of F_2 , because it contains y and

$$(xy)y^{-1} = x.$$

Hence the transformed quotient is trivial. The explicit inverse $\psi(x) = xY, \psi(y) = y$ proves that the map induced by ϕ on the quotients has a two-sided inverse. This is the special case of Theorem 3.4.

Diagnostic: The quotient proof uses normal closure; the later three-move sequence proves the stronger AC statement.

Answers for Section 4: transported moves and bounds

Answer to Exercise 4.1

The original target is xr_1X . Applying ϕ gives

$$\phi(xr_1X) = xy\phi(r_1)YX.$$

Starting with $R = \phi(r_1)$, first conjugate by y , then by x :

$$R \xrightarrow{\text{AC3}} yRY \xrightarrow{\text{AC3}} x(yRY)X = xyRYX.$$

The second relator is unchanged. Thus one original generator conjugation became two source-convention AC3 moves.

Diagnostic: Conjugate by the letters of $\phi(x) = xy$ in reverse order: first y , then x .

Answer to Exercise 4.2

Freely reduce the images:

$$\phi(x) = \mathbf{xyY} \longrightarrow \mathbf{x}, \quad \phi(y) = \mathbf{Yx}.$$

This is an automorphism; for example, its inverse is $\psi(x) = x, \psi(y) = xY$, since

$$\psi(Yx) = yXx = y, \quad \phi(xY) = xXy = y,$$

and both compositions also fix x . The image lengths are 1 and 2, so $\Lambda(\phi) = 2$. The sharper sum is

$$10 + 4 + 7 \cdot 1 + 3 \cdot 2 = 14 + 7 + 6 = \boxed{27}.$$

The coarser bound $\Lambda(\phi)N$, with $N = 10 + 4 + 7 + 3 = 24$, would be 48.

Diagnostic: Reduce $\phi(x), \phi(y)$ before measuring them, and use the actual conjugating letters for the sharper sum.

Answer to Exercise 4.3

For AC1,

$$\phi(r_i r_j) = \phi(r_i)\phi(r_j)$$

as a group identity. Any cancellation merely changes the displayed reduced representative after the one AC1 operation. For AC2,

$$\phi(r_i^{-1}) = \phi(r_i)^{-1}.$$

Again free reduction is equality in F_2 , not an extra move. Therefore one transformed operation of the same type reaches the correct free-group element.

Diagnostic: Move count concerns operations on group elements; free reduction chooses their word representatives.

Answer to Exercise 4.4

Under generator-only AC3, $\phi(g)$ may be a word of length greater than one, so the conjugation must be decomposed letter by letter and the correct arrow is AC*. If AC3 is defined to permit an arbitrary conjugating word, the single operation

$$\phi(r_i) \mapsto \phi(g)\phi(r_i)\phi(g)^{-1}$$

is one AC3 macro-move. AC1 and AC2 remain one move in either convention.

Diagnostic: This is a convention issue, not a failure of the homomorphism identity.

Answer to Exercise 4.5

Isomorphic trivial quotient groups only show that both presentations present the group 1. The implication “presents 1” $\Rightarrow$ “is AC-trivial” is exactly the unproved Andrews–Curtis assertion. The correct route is stronger: when one presentation already has an AC path, transport that path and then reduce the terminal basis.

Diagnostic: Identify whether an argument proves a fact about the quotient group or a path in the AC graph.

Answers for Section 5: canonicalization and relabeling**Answer to Exercise 5.1**

The four unswapped maps are self-inverse:

$$(x, y), \quad (X, y), \quad (x, Y), \quad (X, Y).$$

Among the swapped maps,

$$(y, x)^{-1} = (y, x), \quad (Y, X)^{-1} = (Y, X),$$

and

$$(Y, x)^{-1} = (y, X), \quad (y, X)^{-1} = (Y, x).$$

Here (u, v) abbreviates $x \mapsto u, y \mapsto v$. Direct composition on x, y verifies every claim.

Diagnostic: For a swapped signed map, solve for the preimages of x and y ; do not assume every signed permutation is self-inverse.

Answer to Exercise 5.2

First substitute $y \mapsto Y$, remembering $Y \mapsto y$:

$$YYYxxyyX \mapsto yyyxxYYX.$$

Invert by reversing and changing case:

$$yyyxxYYX^{-1} = xyyXXYYY.$$

Move the final three letters to the front:

$$xyyXXYYY \mapsto YYYxyyXX.$$

This is the first relator of 19_50. The direct substituted string is not stored because canonicalization may invert and rotate it.

Diagnostic: A canonical target is a fingerprint, not necessarily the raw substitution output.

Answer to Exercise 5.3

A signed permutation maps each letter to one letter and maps inverse letters to inverse letters. Hence an adjacent inverse pair maps to an adjacent inverse pair, so free reduction commutes. Because lengths and positions are unchanged, mapping a cyclic rotation equals rotating the mapped word by the same position.

For $\phi(x) = xy$, one x -position becomes two positions and cancellation may occur at image boundaries. The same numerical rotation need no longer select the image of the same block.

Diagnostic: The decisive property is length preservation at the letter level, not merely invertibility.

Answer to Exercise 5.4

All eight have the same AC-triviality truth value by Theorem 4.10. Therefore a 200-move proof for one settles all seven others. It does not imply that each has a shortest path of length 200, or even that the mechanically transported path has length 200: conjugating letters and terminal basis corrections may change the elementary count.

Diagnostic: Transfer the existence of a proof, not its optimal path length.

Answers for Section 6: full Aut(F_2)

Answer to Exercise 6.1

An encoded rotation index refers to positions in the current word. The map $x \mapsto xy$ expands every x to two letters and may create free cancellation next to neighboring images. Consequently word length and the boundaries of cyclic blocks change. Nevertheless, the underlying rotation is a conjugation, and its image is conjugation by the image word, so AC-triviality is preserved.

Diagnostic: Numerical rotation indices are representation-dependent; conjugation identities are group-theoretic.

Answer to Exercise 6.2

An arbitrary endomorphism still satisfies the homomorphism identities, but it may send the terminal basis (x, y) to a non-basis. Without invertibility there is no guaranteed path from $(\phi(x), \phi(y))$ back to (x, y) , nor a reverse implication obtained from ϕ^{-1} . Thus the entire AC-triviality equivalence can fail.

Diagnostic: The certificate must establish “basis,” not merely “homomorphism.”

Answer to Exercise 6.3

Equality of canonical keys is reflexive, symmetric, and transitive because ordinary equality is. The substantive statement is that equality of keys is equivalent to equality of Aut-orbits of unordered cyclic-relator pairs, with each relator taken modulo inversion. Whitehead peak reduction supplies minimum length, and minimum-level connectivity proves that every member of one orbit chooses the same lexicographic minimum. That is the step depending on Theorem 6.1.

Diagnostic: Distinguish “equality is an equivalence relation” from “this key is complete for the intended orbit relation.”

Answer to Exercise 6.4

Running one implementation twice repeats the same assumptions and the same possible bugs. Independently written implementations with different data flows can expose implementation-specific errors when their partitions disagree. Agreement is strong engineering evidence, but the claim of completeness still rests on Whitehead’s mathematical theorem; two programs could share a mistaken theorem or the same misunderstood convention.

Diagnostic: Independent computation validates implementation, while a proof validates the mathematical specification.

Answers for Section 7: S-moves**Answer to Exercise 7.1**

Write $u = abcd$ and $v = ef$, so $r = uv$. Then

$$efabcd = vrv^{-1}$$

uses a conjugator of length 2. Also

$$efabcd = u^{-1}ru$$

uses a conjugator of length 4. Both are correct; the suffix conjugator is shorter in this case.

Diagnostic: A cyclic rotation has two obvious conjugation expressions. Choose the shorter only for a bound, not because the other is false.

Answer to Exercise 7.2

There is one AC1. Since $\varepsilon = -1$, there are two AC2 moves. The AC3 count is

$$m_1 + 2m_2 = 2 + 2 \cdot 3 = 8.$$

The total is

$$1 + 2 + 8 = \boxed{11}.$$

Diagnostic: The other relator must be conjugated back, which is why m_2 occurs twice.

Answer to Exercise 7.3

Starting from (r_i, r_j) :

$$\begin{aligned}
 (r_i, r_j) &\xrightarrow{\text{AC2}} (r_i, r_j^{-1}) \\
 &\xrightarrow{\text{AC3}} (xr_i X, r_j^{-1}) \\
 &\xrightarrow{\text{AC3}} (xr_i X, Y r_j^{-1} y) \\
 &\xrightarrow{\text{AC1}} (xr_i X Y r_j^{-1} y, Y r_j^{-1} y) \\
 &\xrightarrow{\text{AC3}} (xr_i X Y r_j^{-1} y, r_j^{-1}) \\
 &\xrightarrow{\text{AC2}} (xr_i X Y r_j^{-1} y, r_j).
 \end{aligned}$$

The fifth line conjugates the other relator by $y = c_2^{-1}$:

$$y(Y r_j^{-1} y)Y = r_j^{-1}.$$

The target retains the rotated inverse factor created before restoration.

Diagnostic: Restoring the other relator must not alter the product already stored in the target.

Answer to Exercise 7.4

Using $\phi(X) = YX$ and $\phi(Y) = Y$:

$$\phi(c_1) = \phi(Xy) = YXy,$$

which is reduced and has length 3. Also

$$\phi(c_2) = \phi(xY) = xyY \longrightarrow x,$$

which has length 1. For $\varepsilon = +1$, the sharper bound is

$$1 + |\phi(c_1)| + 2|\phi(c_2)| = 1 + 3 + 2 = \boxed{6}.$$

Diagnostic: Using only $\Lambda(\phi)$ would give a valid but weaker bound; reduce the complete conjugator images for the sharper value.

Answers for Section 8: ACA**Answer to Exercise 8.1**

For example,

$$P \xrightarrow{\text{AC1}} P_1 \xrightarrow{\phi} \phi(P_1) \xrightarrow{\text{AC2}} Q.$$

AC1 and AC2 preserve AC-triviality by AC reachability, and the middle step preserves it by Theorem 4.10. Hence $\mathcal{T}(P) = \mathcal{T}(Q)$. The chain does not say that P, Q are adjacent, and the automorphism step need not itself be an ordinary AC path between its endpoints.

Diagnostic: ACA chains preserve a property step by step; they do not collapse every step into one AC edge.

Answer to Exercise 8.2

Every child generated from the chosen representative is produced by a legal S-move and verified automorphism, so every reported edge is sound. However, a different representative $\phi(P)$ of the same orbit may have different word lengths and different encoded rotations, exposing valid children not generated from P . Those missing edges cause incompleteness.

Diagnostic: Soundness asks whether generated edges are valid; completeness asks whether all valid edges are generated.

Answer to Exercise 8.3

Removing the only connecting path leaves two reported components where the full graph has one. Thus the bounded search reports too many components. Selecting one representative from each reported component may duplicate work, but it still selects at least one representative covering every input. This is the safe direction.

Diagnostic: Pruning can split a true component; it cannot fuse two true components.

Answer to Exercise 8.4

A raw AC-graph node is one specific canonical relator pair. A quotient node is the entire Aut(F_2)-orbit of such pairs, represented by its complete Whitehead canonical key. A quotient edge records a legal AC move from the selected orbit representative followed by re-identification of the child orbit.

Diagnostic: Write brackets $[P]_{\text{Aut}}$ when speaking about quotient nodes; this prevents confusing a presentation with its orbit.

Answers for Section 9: the count pipeline**Answer to Exercise 9.1**

$$261 - 93 = 168$$

uses the pure change-of-variable forest edges identifying full Aut-orbits. Then

$$168 - 42 = 126$$

uses the verified ACA search edges in the canonical cap-28 sweep. Finally, the two probe-certified edges join two previously separate singleton pairs:

$$126 - 2 = 124.$$

Every subtraction is a successful union between two formerly distinct components, not merely a listed edge.

Diagnostic: An edge lowers the component count only when its endpoints were previously in different components.

Answer to Exercise 9.2

The two edges internal to existing components lower the count by zero. The other three can lower it by at most one each, so the component count can decrease by at most 3. It may decrease by less if those three become redundant after earlier new unions.

Diagnostic: Count successful unions, not raw edges.

Answer to Exercise 9.3

The bounded graph can omit a valid edge between any two reported components. In the extreme case, all 124 components could belong to one true ACA component; this is compatible with the current search because absence of an edge is not a proof. Therefore 124 is not a lower bound. The proved statement is that the true number is at most 124.

Diagnostic: A search-derived partition of a subgraph refines the partition of the full graph.

Answer to Exercise 9.4

For each of the 20 trivialized representatives, every original input in its certified component is proved AC-trivial. The exact number of settled rows is the sum of the sizes of those 20 components, which cannot be deduced from the number 20 alone. Nothing about the remaining 104 representatives is proved by a failed or unfinished search.

Diagnostic: Report solved components and their member counts, not simply the number of representative runs.

Answer to Exercise 9.5

Repeated discovery of the same two mergers under different ceilings and resource trajectories is evidence of reproducibility. Exactness would require proving that no further connecting path exists. Because every queue terminated on a state, memory, or time cap rather than exhaustion, unvisited states and possible mergers remain. Agreement cannot replace exhaustion or a separating invariant.

Diagnostic: Repeated bounded negative results improve empirical confidence but do not prove a global absence.

Answers for Section 10: certificates**Answer to Exercise 10.1**

Every listed edge may be valid while a separate partition-building bug places an additional, unconnected vertex into a component. Per-edge verification never examines that unsupported assignment. Rebuilding the connected components using only verified edges checks that each reported membership has an actual certificate chain.

Diagnostic: Edge validity and transitive-closure correctness are separate claims and need separate tests.

Answer to Exercise 10.2

Let the rows of M be the exponent-sum vectors of r_1, r_2 .

- AC1 replaces row i by row i plus row j , an elementary unimodular row operation, so the determinant is unchanged.
- AC2 negates one row, so the determinant changes sign and its absolute value is unchanged.
- AC3 sends r_i to $gr_i g^{-1}$. The exponent contributions of g and g^{-1} cancel, so row i is unchanged.

Thus $|\det M|$ is invariant under AC1–AC3. A free-basis automorphism acts on exponent sums by a matrix in $\text{GL}(2, \mathbb{Z})$, i.e. a unimodular column operation, so the absolute determinant is ACA-invariant as well.

Diagnostic: Conjugation is invisible after abelianization because its prefix and suffix have opposite exponent vectors.

Answer to Exercise 10.3

Constancy at 1 means a claimed edge joining inputs with different values would be definitely wrong, so the invariant is useful as an independent consistency check. But because every input already has the same value, it partitions the dataset into only one invariant class and cannot prove any pair inequivalent.

Diagnostic: An invariant detects differences only when it actually takes different values on the objects being compared.

Answer to Exercise 10.4

One example is to change the recorded meeting state on one side while leaving every path step unchanged. Literal replay should end at the original state and fail the equality check against the corrupted claim. Another example is to delete a forest edge but leave both endpoints in the same reported class; the partition rebuild should reject the unsupported membership.

Diagnostic: A good mutation targets one advertised verifier obligation and predicts the exact failed assertion.

Answers for Section 11: oral rehearsal**Answer to Exercise 11.1**

A model two-minute response is:

Because F_2 is free, specifying $x \mapsto xy$ and $y \mapsto y$ defines a homomorphism. It is invertible: the inverse sends $x \mapsto xY$ and $y \mapsto y$, and both compositions fix x, y . Now take an AC path from P to (x, y) . Applying the automorphism to every relator sends AC1 and AC2 to one move of the same type. A generator-conjugation AC3

becomes conjugation by the finite word $\phi(g)$, which expands into finitely many generator conjugations. The transformed path ends at (xy, y) , a free basis, and a finite Nielsen/AC sequence returns it to (x, y) . The inverse automorphism gives the converse. Therefore P is AC-trivial exactly when $\phi(P)$ is.

Diagnostic: The four indispensable points are homomorphism, inverse, transported moves, and terminal basis correction.

Answer to Exercise 11.2

A model response is:

Every one of the 261 inputs lies in one of 124 components connected by verified ACA edges. Each edge preserves AC-triviality, so one representative per component covers every input. The search is incomplete, so additional valid edges may merge components; therefore 124 is an upper bound and the true number may be smaller. Missing edges cause redundant work, not missing coverage.

Diagnostic: Use both words “verified” and “upper bound.”

Answer to Exercise 11.3

A rigorous response is:

I am not claiming the endpoints of every discarded run are connected by a raw AC path. I am using the weaker property actually needed for the experiment: every certified AC or Aut step preserves whether the presentation is AC-trivial. Along a verified ACA chain, the truth value is constant. Therefore if I solve the retained representative, all discarded members of that component follow. The 137 successful union edges reduce 261 vertices to 124 certified coverage components.

Diagnostic: The justification is invariance of the question, not equality of raw AC paths.

Answer to Exercise 11.4

A correct response is:

No. This analysis trivialized zero of the 261 presentations. It grouped the inputs into at most 124 distinct ACA problems. The conditional result is that if one representative from each of those 124 components is later proved AC-trivial, all 261 presentations will be proved AC-trivial.

Diagnostic: Separate classification from solution. Never turn a component count into a trivialization count.

13 Sources and reproducibility map

Purpose	Repository source
Source-paper AC1–AC3	<code>literature/txt/math_ml_paper_2408.15332.txt</code> , lines 291–302
Master 261-to-124 account	<code>results/equivalence_classes/EQUIVALENCE_FINDING.md</code>
Current 124-class proof book	<code>results/equivalence_classes/PROOFS.md</code>
Current 137-edge machine certificates	<code>results/equivalence_classes/certificates.json</code>
Word algebra and S-move replay	<code>experiments/equivalence_classes/lib/words.py</code>
S-move generation	<code>experiments/equivalence_classes/lib/acmoves.py</code>
Whitehead canonicalization with witness	<code>experiments/equivalence_classes/lib/autcanon.py</code>
Independent Whitehead implementation	<code>experiments/analysis/whitehead.py</code>
ACA quotient search	<code>experiments/equivalence_classes/search/aut_search.py</code>
Base certificate verifier	<code>experiments/equivalence_classes/verify/verify_proofs.py</code>
125/124 probe verifier	<code>experiments/equivalence_classes/verify/verify_probe_merges.py</code>
Overnight 124 artifacts	<code>results/equivalence_classes/probe/probe_overnight_seam*.json</code>

Classical theorem dependencies

- The universal property of free groups, proved in Theorem 2.2.
- Nielsen’s theorem on bases of finitely generated free groups, used to certify automorphisms, together with Nielsen generation (Theorem 4.8) to reduce the terminal basis.
- Whitehead’s peak-reduction theorem, stated in Theorem 6.1, used to make the 168 Aut-orbit count exact.
- The Andrews–Curtis conjecture, used only to explain why presenting the trivial group does not currently imply a known AC trivialization.